

EQ2330 Image and Video Processing

Project 1

Chong, Ngai Chun Adrian
ncachong@kth.se

Oğuzhan Oğuz
oguzhano@kth.se

November 21, 2025

Summary

This study investigates fundamental spatial and frequency domain image enhancement techniques such as specifically contrast enhancement, noise reduction, and image restoration. Histogram equalization was applied to a low-contrast simulation. It is successfully expanding dynamic range while demonstrating the limitations of discrete intensity distribution. In denoising experiments, mean and median filters were evaluated. Results indicate that the median filter is superior for removing salt-and-pepper noise while preserving edges. Finally, a frequency-domain Wiener filter was applied to remove out-of-focus blur. By approximating the signal-to-noise ratio, the filter effectively recovered fine details despite small ringing artifacts caused by boundary discontinuities.

1 Introduction

This project aims to investigate different image enhancement techniques when applied to degraded images. Both spatial and frequency domain processing algorithms are discussed. Throughout the entire report, the algorithms are tested using the provided “Lena” image as shown in Figure 1.

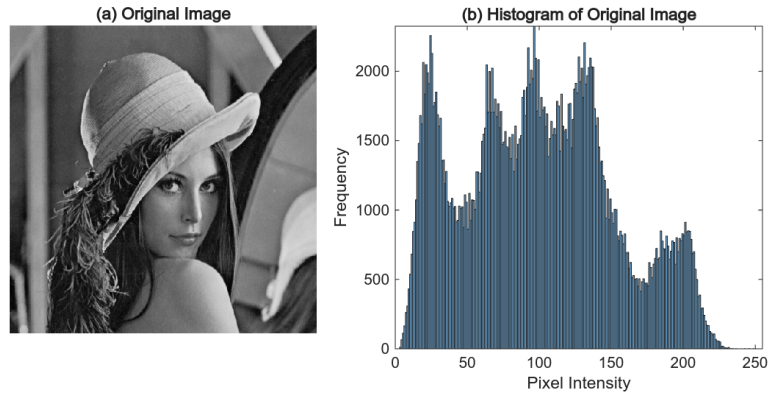


Figure 1: The original “Lena” test image and its histogram.

The report is separated into three main sections: section 2 discusses applying histogram equalization to images with reduced contrast, section 3 explores filtering techniques for removing gaussian and salt-and-pepper noise, and section 4 applies Wiener filtering to deblur out-of-focus images.

2 Histogram equalization

This section describes the process of simulating a low-contrast image using the “Lena” test image and discusses the effects of applying histogram equalization on the image.

2.1 Simulating low-contrast image

Starting from an ideal image, we could simulate a low contrast image by reducing its dynamic range using the point-wise transformation

$$g_c(x, y) = \min(\max(\lfloor a \cdot f(x, y) + b \rfloor, 0), 255) \quad (1)$$

where we choose $a = 0.2$ and $b = 50$. Applying the transformation in (1) on the original “Lena” image in Figure 1(a) results in

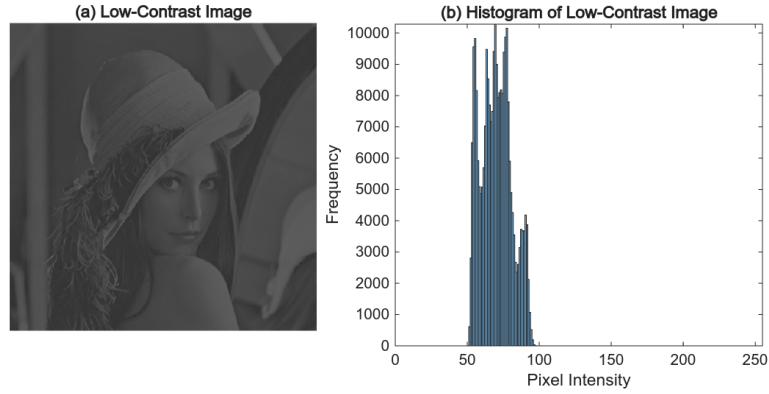


Figure 2: The degraded “Lena” image with low-contrast and its histogram.

Comparing the histograms of the degraded image in Figure 2(b) to the original in Figure 1(b), we can observe that the general shape of the histogram remained the same while the distribution is squished and translated. This matches our intuition as each pixel value is scaled and offset uniformly over the entire image by (1).

2.2 Applying histogram equalization

Described in section 3.3.1 of [1], histogram equalization is implemented using the point-wise transformation

$$s_k = T(r_k) = \frac{1}{n} \sum_{i=0}^k n_i, \quad k = 0, 1, \dots, L-1 \quad (2)$$

where s_k is the new value for pixels with original value r_k , n_k is the number of pixels with value r_k , and n is the total number of pixels.

To implement the transformation in (2) for 8-bit grayscale images, we can use `histcounts` and `cumsum` to generate a lookup table of new pixel values for all 256 intensity levels. Since the new intensity values are normalized to $[0, 1]$ for (2), the output image would also need to be scaled by 255 and rounded to maintain the same format.

2.3 Effects of histogram equalization

Applying histogram equalization to Figure 2(a) produces results in Figure 3. We note that the resultant histogram is not completely flat as in the theoretical continuous case. Under histogram equalization, multiple original intensity levels may be mapped to the same output level due to rounding, while no original level can be split across different output levels. As a result, some output bins receive many pixels, others remain empty. Therefore, for a general discrete digital image, it is not guaranteed to produce a flat histogram.

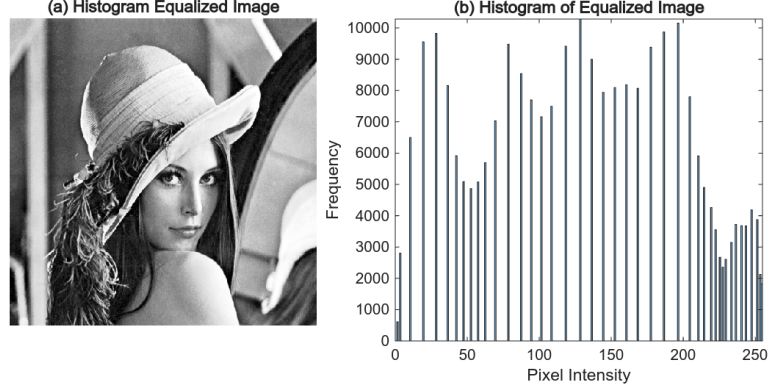


Figure 3: Resultant image after applying histogram equalization to the contrast-reduced image and its histogram.

3 Image Denoising with filtering

This section investigates the denoising effects of spatial smoothing and order-statistic filters on additive gaussian noise and salt-and-pepper noise. In particular, a 3×3 mean filter and 3×3 median filter are used.

Starting from the original “Lena” image in Figure 1(a), a zero-mean gaussian noise with variance 64 and salt-and-pepper noise with $p(0) = p(255) = .05$ are applied, resulting in the noisy images found in Figure 4.

Comparing the histograms to the original, we can observe that Figure 4(b) has more rounded peaks while maintaining a similar distribution; as for Figure 4(d), it is dominated by two peaks at 0 and 255 respectively while the center distribution remains the same.

3.1 Mean filter

To apply a 3×3 mean filter on the noisy images, a convolution with the kernel matrix

$$K = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad (3)$$

is used. To address the boundaries, the image is zero-padded (the default behavior for the Matlab `conv2` function). The results are found in Figure 5.

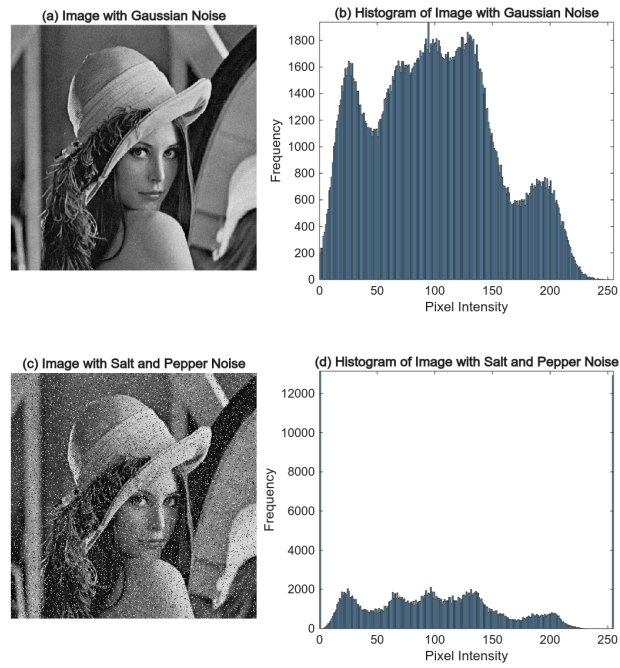


Figure 4: Images with gaussian and salt-and-pepper noise and their histogram.

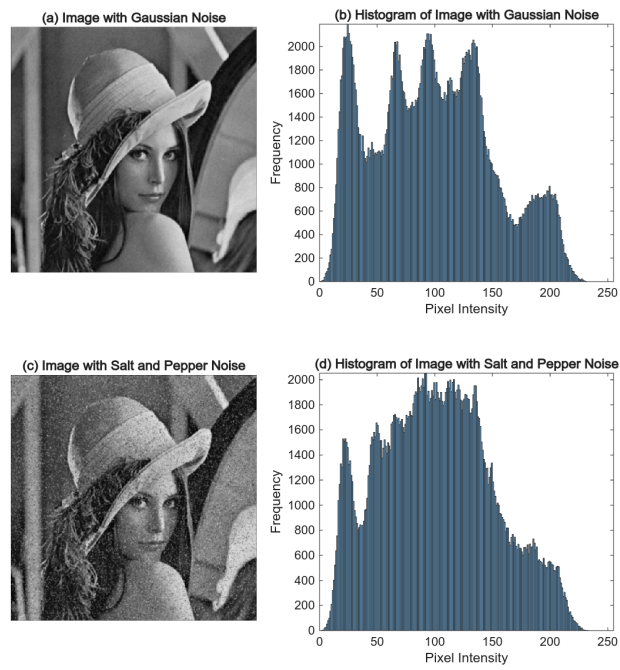


Figure 5: Mean-filtered noisy images with their histograms.

3.2 Median filter

Since a 3×3 median filter is a non-linear transformation, it cannot be represented as a convolution of a kernel matrix. Therefore, we would have to apply the median filter manually: for each pixel $f[x, y]$, update with the median value of its 3×3 neighbor. The boundaries are handled by zero-padding to obtain enough values. In Matlab, this operation can be done using the `medfilt2` function. The results are found in Figure 6.

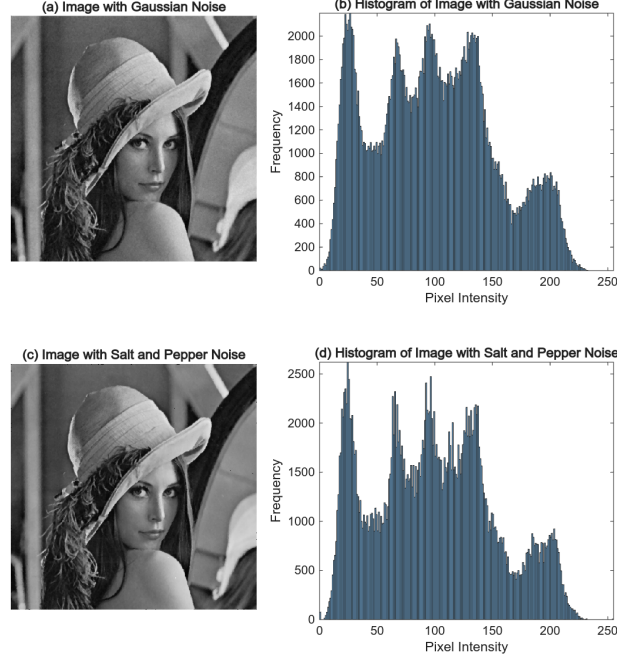


Figure 6: Median-filtered noisy images with their histograms.

3.3 Comparison

For gaussian noise, it is observed that both the mean and median filters are effective in removing the noise. As additive gaussian noise are approximately white, adding random fluctuations, both filters acts as low-pass filters to suppress these local high-frequency fluctuations. However, the same cannot be said for salt-and-pepper noise, despite also adding high-frequency components.

Applying the mean filter on the image with salt-and-pepper noise fails to remove the noise. Since salt-and-pepper noise creates extreme values near unmodified pixels, by the nature of mean filters, these extreme values (0 or 255) heavily influence the averaged result in their local neighborhood. Even a single noisy pixel inside the filter window pulls the output intensity significantly toward 0 or 255, causing additional noise. On the other hand, since median filters are less sensitive to outliers, it is effective in removing salt-and-pepper noise.

In addition, median filtering appears to better preserve edges found in the image. Mean-filtering fails as it performs smoothing over the entire image, softening existing edges.

4 Frequency domain filtering

In this section, image degradation and restoration in the frequency domain are investigated. The blurred and quantized image $g(x, y)$ is modeled by the standard degradation model

$$g(x, y) = (h * f)(x, y) + \eta(x, y), \quad (4)$$

where f is the original image, h is the point spread function (PSF) of the out-of-focus blur and η is the quantization noise introduced by the 8-bit representation.

The blurred image is generated by convolving the original image with a Gaussian blur kernel produced by `myblurgen('gaussian', r)` with radius $r = 8$. In the implementation this corresponds to using `conv2(f, h, 'same')`. The resulting out-of-focus image is then quantized to 8-bit integer values, which introduces the noise term $\eta(x, y)$ in (4).

Applying the two-dimensional DFT to (4) gives

$$G(u, v) = H(u, v)F(u, v) + N(u, v), \quad (5)$$

where F , G and H are the DFTs of f , g and h , and N is the transform of the noise. The DFT implicitly assumes that the input image is *periodic*. Consequently, the Convolution Theorem dictates that multiplication in the frequency domain corresponds to *circular* convolution in the spatial domain. However, the physical blur represents a *linear* convolution. If the image and the kernel are transformed without padding, the circular nature of the DFT causes opposite edges of the image to interfere, resulting in severe wraparound errors.

To approximate linear convolution and avoid spatial aliasing, both the image and the PSF are zero-padded to a size $P \times Q$ with $P \geq M + L_h - 1$ and $Q \geq N + L_w - 1$, where $M \times N$ is the image size and $L_h \times L_w$ is the PSF size. This choice prevents wraparound but introduces sharp transitions from the image to zero at the extended borders. These transitions are responsible for the visible boundary artefacts and for the strong vertical and horizontal lines in some spectra.

To visualise the effect of the degradation on the frequency content, the two-dimensional DFT of the original image and of the degraded image is computed. The spectra are centered with `fftshift` and displayed as log-magnitude images $\log(1 + |F(u, v)|)$ and $\log(1 + |G(u, v)|)$.

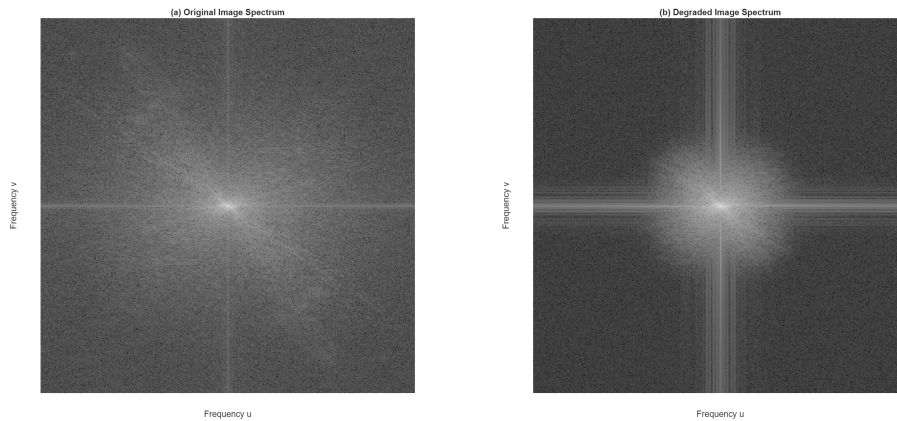


Figure 7: Log-magnitude spectra of the original image and the degraded image.

Figure 7 confirms that the blur behaves as a low-pass filter. Fine details

present in the original image are attenuated in the degraded spectrum, while the low-frequency region remains dominant.

For restoration, a Wiener filter is designed in the frequency domain. The objective is to find a linear estimator $\hat{F}(u, v)$ of $F(u, v)$ that minimizes the mean-squared error $\mathbb{E}\{|F(u, v) - \hat{F}(u, v)|^2\}$. Under the standard assumptions that F and N are uncorrelated, zero-mean random fields with power spectral densities $S_F(u, v)$ and $S_N(u, v)$, the optimal estimator is

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + K(u, v)} G(u, v), \quad K(u, v) = \frac{S_N(u, v)}{S_F(u, v)}, \quad (6)$$

where $(\cdot)^*$ denotes complex conjugation. In the implementation a common simplification is adopted, namely that $K(u, v)$ is spatially constant and can be approximated by the ratio of noise and image variances,

$$K_0 \approx \frac{\sigma_\eta^2}{\sigma_g^2}, \quad (7)$$

where σ_η^2 is the variance of the quantization noise and σ_g^2 is the variance of the degraded image. To compensate for modeling errors and to trade off sharpness versus noise amplification, the final parameter is chosen as

$$K = \alpha K_0, \quad \alpha = 50, \quad (8)$$

which provides a good compromise for the considered test images. The restored image is then obtained by the inverse DFT

$$\hat{f}(x, y) = \mathcal{F}^{-1}\{\hat{F}(u, v)\}, \quad (9)$$

followed by cropping back to the original size $M \times N$.

The Wiener filter in (6)–(8) is applied to the degraded images using the known PSF h and the noise variance σ_η^2 estimated from the difference between the blurred image and its 8-bit version. The same procedure is used for all test images.

Visually, the restored images in Figures 8–9 are significantly sharper than the degraded ones, and most of the blur is removed without excessive amplification of quantization noise. The remaining ringing and boundary artefacts are consistent with the theoretical properties of the Wiener filter and with the finite-support DFT implementation based on zero-padding.

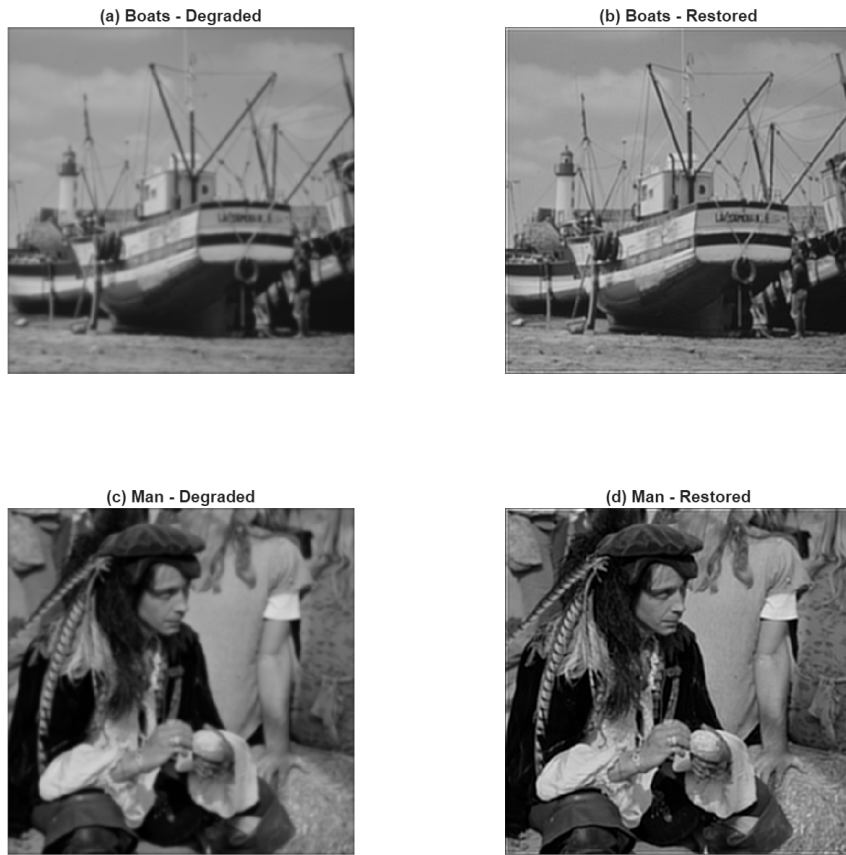


Figure 8: Deblurring results on two test images ("Boats" and "Man").



Figure 9: Deblurring of the "Lena" image.

5 Conclusions

In this project, three types of image enhancement algorithms were implemented. First, histogram equalization was analyzed. This method is useful for redistributing brightness. However, the process is limited by the discrete quantization levels. As a result, getting a perfectly uniform distribution is not mathematically possible for digital images. Next, noise reduction techniques were tested. The results indicate that the best filter depends on the noise type. Linear mean filters reduce Gaussian noise but do not work well against impulsive disturbances. On the other hand, the non-linear median filter successfully removed salt-and-pepper noise. Unlike the linear options, this method kept the image sharp without blurring the edges. The last task involved using a Wiener filter in the frequency domain. This technique worked well to deblur the out-of-focus images. By using the noise-to-signal ratio, the filter prevented too much noise amplification. Zero-padding was also used to stop spatial aliasing. But this creates sharp changes at the borders. These border issues caused some ringing artifacts to appear in the restored image.

Appendix

Who Did What

Chong completed the code development and reporting for sections concerning histogram equalization and image denoising. Oguz completed the code development and reporting for sections concerning frequency domain filtering. After the project was completed, the two authors reviewed the entire project and provided the necessary feedback.

References

- [1] Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, Prentice Hall, 2nd ed., 2002